

1. State True or False. Structures in C++ can contain member functions.

Answer: True

2. What is the output of the following program?

```
#include <iostream>
using namespace std;

struct Point {
    int x, y;

    void set(int a, int b) { x = a; y = b; }
    void print() { cout << x << " " << y; }
};

int main() {
    Point p;
    p.set(3, 7);

    p.print();
    return 0;
}
```

Answer: 3 7

3. Which class is used to open a file for reading only?

- a) ofstream
- b) fstream
- c) ifstream
- d) iostream

Answer: c)

4. What is the difference between `ios::app` and `ios::ate`?

Answer: In `ios::app` every write goes to end of file, even if pointer is moved while in `ios::ate` pointer starts at end, but writes can occur anywhere after repositioning.

5. Streams in C++ are device dependent.

Answer: False

6. State True or False. A stream state method `good()` returns true only when none of the eofbit, failbit, or badbit flags are set.

Answer: True

7. Write a C++ statement that will declare and open a binary file `picture.gif` for input.

Answer: `ifstream ifs ("picture.gif", ios::binary);`

8. What do the following statements do?

```
ofstream stream;  
stream.open("data.txt");
```

- a) Opens the file data.txt for input.
- b) Opens the file data.txt for output, the statement fails if the file already exists.
- c) Opens the file data.txt for output, the contents of the file are destroyed if the file already exists.
- d) Opens the file data.txt for input, the statement fails if the file does not exist.

Answer: c)

9. In C++, which member function is used to flush an output stream buffer?

Answer: `ostream::flush()`

10. What is the key difference between `get(char* s, n)` and `getline(char* s, n)` when reading from a file?

- a) `get()` discards the delimiter; `getline()` leaves it in the stream
- b) `getline()` discards the delimiter; `get()` leaves it in the stream
- c) Both discard the delimiter
- d) Both leave the delimiter in the stream

Answer: b)

11. Write a statement to create a dynamic array of integers of size 5 using new operator in C++?

Answer: `int *p = new int[5];`

12. Write a statement to create a dynamic array of pointers (to) integers of size 5 using new operator in C++?

Answer: `int **p = new int *[5];`

13. State True or False. While both new operator and `malloc` function can be used for dynamic memory allocation, the new operator calls the constructor but `malloc` does not.

Answer: True

14. Given a class A as defined below:

```
class A{  
    int x;  
    int y;  
public:  
    void disp();  
}
```

Define a destructor (within the class) that displays the message "Object destroyed"

Answer:

```
~A(){cout<<"Object destroyed";}
```

15. State True or False. A destructor is called when the block that declares an object of a class ends.

Answer: True

16. What is the output of the following program?

```
#include <iostream>
using namespace std;
struct Counter {
    Counter() { cout << "C "; }
    ~Counter() { cout << "D "; }
};

int main() {
    Counter a;
    Counter b;
    return 0;
}
```

Answer: C C D D

17. In which order are constructors invoked when creating a derived class object?

- a) Derived constructor first, then base constructor
- b) Base constructor first, then derived constructor
- c) Only the derived constructor is called
- d) Order is determined by the compiler

Answer: b)

18. State True or False. Members in a constructor initializer list are initialized in the order they appear in the initializer list.

Answer: False (members are initialized in their declaration order inside the class, regardless of initializer list order)

19. State True or False. `std::list` supports random access using operator `[]`.

Answer: False

20. State True or False. In C++, the declaration `int& y = x;` creates `y` as a separate copy of `x`.

Answer: False

21. State True or False. Given `it` as an iterator for `std::list<int>` the expression `it+ 2` is valid.

Answer: False

22. What is the output of the following program.

```
#include <iostream>
using namespace std;

struct Vals {
    int b;    // declared first
    int a;    // declared second
    Vals(int x) : a(x), b(a * 2) {}
};

int main() {
    Vals v(5);
    cout << v.a << " " << v.b;
    return 0;
}
```

Answer: b)

23. What is the output of the following program?

```
#include <iostream>
using namespace std;
class Animal {
public:
    Animal() { cout << "Animal "; }
    ~Animal() { cout << "~Animal"; }
};
class Dog : public Animal {
public:
    Dog() { cout << "Dog "; }
    ~Dog() { cout << "~Dog "; }
};
int main() {
    Dog d;
    return 0;
}
```

Answer: Animal Dog ~Dog ~Animal

24. Given `v` as a `std::vector<int>`, what does `v.at(i)` do differently from `v[i]`?

- a) Returns the element faster
- b) Throws `std::out_of_range` if `i` is out of bounds
- c) Returns a pointer to the element
- d) Inserts an element at position `i`

Answer: b)

25. What is the output of the following program?

```
#include <iostream>
#include <vector>

using namespace std;

int main() {
    vector<int> v;

    v.push_back(10);
    v.push_back(20);
    v.push_back(30);

    for (auto x : v)
        cout << x << " ";
    return 0;
}
```

Answer: 10 20 30

26. Which STL container is most suitable for frequent insertions and deletions in the middle of a sequence?

Answer: `std::vector`

27. What does the capacity of a vector refer to in C++?

- a) The number of elements currently stored in the vector
- b) The maximum number of elements that can be stored in the vector without the need for reallocation
- c) The maximum size to which the vector can grow due to implementation limitations
- d) The unused memory space of the vector

Answer: b)

28. Which of the following statements declares a vector of integer values.

- a) `std::vector<integer> vecOfInts;`
- b) `std::vector<int> vecOfInts;`
- c) `std::vector(int) vecOfInts;`
- d) `std::vector<int> vecOfInts;`

Answer: d)

29. What is true about this statement in C++: `std::vector<int> vecInts(5);`

- a) Initializes a Vector with 5 integer values, each set to default value of 0.
- b) Initializes a Vector with 5 integer values.
- c) Initializes a Vector with 5 integer values, each set to garbage.
- d) Initializes a Vector with 5 integer values, each set to default value of 1.

Answer: a)

30. Which STL container is most suitable for frequent insertions and deletions in the middle of a sequence?
- a) `std::vector`
 - b) `std::array`
 - c) `std::list`
 - d) `std::deque`

Answer: c)